

$$G_X(t) = ct(1+t)^5,$$

(a) Find the value of c .

[1]

[2]



The random variable Y is the sum of two independent values of X .

- (c) Write down the probability generating function of Y and hence find $\text{Var}(Y)$. [4]

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- (d) Find $P(Y = 5)$. [2]

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